Subareas and Interdisciplinary Areas in Reinforcement Learning

Below is a curated list of key subareas in Reinforcement Learning (RL), including both core methodological areas and interdisciplinary applications. Each subarea includes a brief summary, followed by 3 selected papers (a mix of cornerstone classics and recent works). The list draws from established RL frameworks, such as those involving model-based planning, hierarchy, multi-agent systems, and learning from demonstrations or feedback. Interdisciplinary connections include robotics, games, conversational AI, image generation, recommendataion, economics, etc.

1. Model-Based Reinforcement Learning

Involves learning a dynamics model of the environment to simulate trajectories for planning, improving sample efficiency over model-free methods.

- **When to Trust Your Model: Model-Based Policy Optimization** (Janner et al., 2019): Classic work emphasizing uncertainty-aware model usage. arXiv
- **MOReL: Model-Based Offline Reinforcement Learning** (Kidambi et al., 2020): Introducing MOReL, an algorithmic framework for model-based offline RL that learns a pessimistic MDP to handle distribution shift in offline settings. NeurIPS
- **Model-based reinforcement learning for ultrasound-driven microrobots** (Lightner et al., 2025): Recent work applying model-based RL to control microrobots via imagined environments. bioRxiv

2. Theoretical Foundations of Reinforcement Learning

Focuses on mathematical underpinnings, including convergence guarantees, regret bounds, and sample complexity in RL algorithms, providing rigor for practical methods.

- **Reinforcement Learning: A Survey** (Kaelbling et al., 1996): Cornerstone survey outlining core theoretical concepts like Markov decision processes and value iteration. JAIR
- **Regret Analysis of Stochastic and Nonstochastic Multi-armed Bandit Problems** (Bubeck & Cesa-Bianchi, 2012): Classic on regret minimization in bandits, foundational for RL theory. arXiv
- **Reinforcement Learning: An Overview** (Murphy, 2024): Recent overview of theoretical advances, including up-to-date discussions on convergence in deep RL settings. arXiv

3. Exploration in Reinforcement Learning

Addresses the exploration-exploitation tradeoff, developing strategies like epsilon-greedy or intrinsic motivation to efficiently discover optimal policies in unknown environments.

- **Efficient Exploration in Reinforcement Learning** (Thrun, 1992): Cornerstone on fundamental exploration methods, including directed and undirected strategies. PDF
- **Intrinsic Motivation Systems for Autonomous Mental Development** (Oudeyer et al., 2007): Classic introducing curiosity-driven exploration via prediction errors. PDF
- **The Role of Inherent Bellman Error in Offline Reinforcement Learning with Linear Function Approximation** (Golowich & Moitra, 2024): Recent theoretical analysis of exploration errors in offline settings, with implications for regret bounds. arXiv

4. Multi-Agent Reinforcement Learning (MARL)

Extends RL to scenarios with multiple interacting agents, addressing cooperation, competition, and communication in games.

- **Multi-Agent Actor-Critic for Mixed Cooperative-Competitive Environments** (Lowe et al., 2017): Cornerstone on MADDPG, enabling centralized training with decentralized execution. arXiv
- **Value-Decomposition Networks For Cooperative Multi-Agent Learning** (Rashid et al., 2018): Classic QMIX paper for value factorization in cooperative MARL. arXiv
- **A Comprehensive Review of Multi-Agent Reinforcement Learning in Video Games** (Li et al., 2025): Recent review on MARL applied to video games. arXiv

5. Hierarchical Reinforcement Learning

Decomposes complex tasks into hierarchies of sub-policies or options for better long-horizon planning and transfer.

- **Between MDPs and semi-MDPs: A framework for temporal abstraction in reinforcement learning** (Sutton et al., 1999): Cornerstone introducing the options framework for hierarchy. PDF
- **Feudal Reinforcement Learning** (Dayan & Hinton, 1993): Classic on multi-level hierarchies with manager-worker structures. PDF
- **Large language model assisted hierarchical reinforcement learning for pathfinding** (Li et al., 2025): Recent integration of LLMs with hierarchical RL for complex navigation. ResearchGate

6. Inverse Reinforcement Learning (IRL)

Infers reward functions from expert demonstrations, useful for robotics and autonomous driving.

- **Algorithms for Inverse Reinforcement Learning** (Ng & Russell, 2000): Cornerstone formalizing IRL as reward recovery from behavior. PDF
- **Apprenticeship Learning via Inverse Reinforcement Learning** (Abbeel & Ng, 2004): Classic applying IRL to match expert trajectories. PDF
- **Inverse Reinforcement Learning Meets Large Language Model Post-Training: Basics, Advances, and Opportunities** (Sun & Schaar, 2025): Recent review linking IRL to LLM alignment. arXiv

7. Imitation Learning

Learns policies directly from demonstrations without explicit rewards, bridging RL and supervised learning for tasks like manipulation.

- **A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning** (Ross et al., 2011): Cornerstone on DAGGER, an interactive imitation method. PDF
- **Generative Adversarial Imitation Learning** (Ho & Ermon, 2016): Classic GAIL using adversarial training for imitation. arXiv
- **A Survey on Imitation Learning for Contact-Rich Tasks in Robotics** (Tsuji et al., 2025): Recent survey on IL for physical interactions in robotics. arXiv

8. Offline Reinforcement Learning

Learns policies from fixed datasets without further interaction, addressing data efficiency in real-world scenarios like healthcare.

- **Conservative Q-Learning for Offline Reinforcement Learning** (Kumar et al., 2020): Cornerstone on CQL to prevent overestimation in offline settings. arXiv
- **Offline Reinforcement Learning with Realizability and Single-policy Concentrability** (Zhan et al., 2022): Classic on theoretical guarantees for offline RL. arXiv
- **Offline reinforcement learning for job-shop scheduling problems** (Echeverria et al., 2025): Recent application to combinatorial optimization. arXiv

9. Safe Reinforcement Learning

Incorporates safety constraints to avoid harmful actions, critical for robotics and autonomous systems.

- **A Comprehensive Survey on Safe Reinforcement Learning** (Garcia & Fernandez, 2015): Cornerstone survey on safe exploration methods. JMLR
- **Constrained Policy Optimization** (Achiam et al., 2017): Classic on CPO for constraint satisfaction in RL. arXiv
- **Offline Safe Reinforcement Learning Using Trajectory Classification** (Gong et al., 2025): Recent offline approach classifying safe trajectories. arXiv

10. Reinforcement Learning from Human Feedback (RLHF)

Aligns RL agents with human preferences via feedback, widely used in LLMs for value alignment.

- **Deep Reinforcement Learning from Human Preferences** (Christiano et al., 2017): Cornerstone introducing RLHF for complex tasks. arXiv
- **Fine-Tuning Language Models from Human Preferences** (Ziegler et al., 2019): Classic application to language models. arXiv
- **M^3HF: Multi-agent Reinforcement Learning from Multi-phase Human Feedback of Mixed Quality** (Wang et al., 2025): Recent multi-agent extension with phased feedback. ICML

11. Meta-Reinforcement Learning

Enables fast adaptation to new tasks by learning to learn, drawing from meta-learning techniques.

- **Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks** (Finn et al., 2017): Cornerstone MAML applied to RL for quick adaptation. arXiv
- **RL^2: Fast Reinforcement Learning via Slow Reinforcement Learning** (Duan et al., 2016): Classic on recurrent policies for meta-RL. arXiv
- **Meta-Reinforcement Learning with Adaptation from Human Feedback via Preference-Order-Preserving Task Embedding** (Xu & Zhu, 2025): Recent work proposing POEM that enables few-shot policy adaptation from human preferences by learning a task embedding space that preserves preference order. OpenReview

12. Reinforcement Learning for Large Language Models

This subarea focuses on RL methods for fine-tuning LLMs, including new variants like Group Relative Policy Optimization (GRPO) and preference-based techniques, to enhance reasoning, alignment, and performance beyond supervised fine-tuning.

- **Direct Preference Optimization: Your Language Model is Secretly a Reward Model** (Rafailov et al., 2023): Introduces DPO, a simplified alternative to RLHF that directly optimizes language models on preference data without an explicit reward model, enabling efficient alignment. arXiv
- **DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Pure Reinforcement Learning** (DeepSeek-AI Team, 2025): Demonstrates that reasoning in LLMs can be boosted through pure RL without supervised fine-tuning, achieving performance comparable to leading models on math and code tasks. arXiv
- **DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models** (DeepSeek-AI Team, 2024): Introduces Group Relative Policy Optimization (GRPO), a PPO variant for RL fine-tuning of generative LLMs, enhancing reasoning in text-based math tasks. arXiv

13. Reinforcement Learning for Image Generation

This subarea applies RL to optimize diffusion and other generative models for image synthesis, improving alignment, stability, and resistance to attacks like jailbreaking.

- **Diffusion Model Alignment Using Direct Preference Optimization** (Wallace et al., 2023): Applies RL to refine diffusion-based image generation models. arXiv
- **AR-GRPO: Training Autoregressive Image Generation Models via Reinforcement Learning** (Yuan et al., 2025): Integrates online RL with GRPO into autoregressive models to improve image quality and diversity, inspired by LLM fine-tuning techniques. arXiv
- **Towards Better Alignment: Training Diffusion Models with Reinforcement Learning Against Sparse Rewards** (Hu et al., 2025): Recent CVPR work fine-tuning diffusion models via RL for better prompt following. arXiv

14. Reinforcement Learning for Speech Generation

This subarea leverages RL to fine-tune text-to-speech (TTS) and speech synthesis models, focusing on emotion, quality, and efficiency in diffusion-based architectures.

- **Reinforcement Learning for Emotional Text-to-Speech Synthesis with Improved Emotion Discriminability** (Liu et al., 2021): Leverages RL to improve emotional expressiveness in speech synthesis. arXiv
- **Fine-Tuning Text-to-Speech Diffusion Models Using Reinforcement Learning** (Chen et al., 2025): Recent Interspeech paper applying RL to optimize diffusion TTS for better audio quality and reduced diffusion sampling steps. arXiv
- **Emo-DPO: Controllable Emotional Speech Synthesis through Direct Preference Optimization** (Gao et al., 2025): Recent Interspeech paper using RL to improve emotional expressiveness. arXiv

15. Reinforcement Learning for Robotics

This subarea applies RL to robotic control, manipulation, and planning, often integrating diffusion models or policy optimization for real-world tasks like visuomotor learning and multi-robot coordination.

- **Diffusion Policy: Visuomotor Policy Learning via Action Diffusion** (Chi et al., 2023): Cornerstone paper introducing Diffusion Policy, representing robot behaviors as conditional denoising diffusion processes for scalable visuomotor control. arXiv
- **Diffusion Policy Policy Optimization** (Ren et al., 2024): Classic extension providing a framework for fine-tuning diffusion-based policies with RL, improving sample efficiency in robotic tasks. arXiv
- **Deep Reinforcement Learning for Robotics: A Survey of Real-World Successes** (Tang et al., 2025): Recent survey evaluating RL's practical applications in robotics, highlighting successes in manipulation, navigation, and multi-agent systems. AAAI

16. Reinforcement Learning in Recommendation Systems

Applies RL to optimize sequential recommendations, modeling user interactions as Markov decision processes to maximize long-term engagement in platforms like e-commerce or streaming.

- **Reinforcement learning based recommender systems: A survey** (Afsar et al., 2021): Cornerstone survey on RL for recommender systems, covering key frameworks and challenges. arXiv
- **Deep Exploration for Recommendation Systems** (Zhu & Roy, 2021): Classic on addressing sparse feedback with contextual bandits in recommendation. arXiv
- **RecoMind: A Reinforcement Learning Framework for Optimizing In-Session User Satisfaction in Recommendation Systems** (Ayed et al., 2025): Recent simulator-based RL for web-scale recommendations, focusing on user retention. arXiv

17. Reinforcement Learning in Operations Research

Uses RL for optimization problems like inventory management, scheduling, and supply chain, where dynamic decisions under uncertainty are key.

- **Deep reinforcement learning for inventory control: A roadmap** (Boute et al., 2022): Cornerstone roadmap for applying DRL to inventory problems in operations. ScienceDirect
- **Reinforcement learning for combinatorial optimization: A survey** (Mazyavkina et al., 2021): A survey of RL applied to combinatorial optimization ScienceDirect
- **Deep reinforcement learning for the real-time inventory rack storage assignment and replenishment problem** (Teck et al., 2025): Recent DRL for warehouse optimization, minimizing robot travel. ScienceDirect

18. Reinforcement Learning in Economics

Integrates RL with economic modeling for policy design, market simulations, and behavioral analysis, often in multi-agent settings for equilibria discovery.

- **Reinforcement Learning in Economics and Finance** (Charpentier et al., 2020): Cornerstone on RL applications in economics, game theory, and operations. arXiv
- **Computational Performance of Deep Reinforcement Learning to Find Nash Equilibria** (Graf et al., 2023): Classic on RL for strategic behavior in price competition. Springer
- **Deep Reinforcement Learning in a Monetary Model** (Chen et al., 2025): Recent DRL for bounded rationality in dynamic stochastic general equilibrium models. arXiv

19. Reinforcement Learning in Social Sciences

Applies RL to model human behavior, social dilemmas, and policy interventions, incorporating elements like cooperation and influence in multi-agent social contexts.

- **Learning from other minds: an optimistic critique of reinforcement learning models of social learning** (Vélez & Gweon, 2021): Cornerstone critique of RL for social information valuation. ScienceDirect
- **A reinforcement learning approach to explore the role of social expectations in altruistic behavior** (Castañón et al., 2023): Classic agent-based model using RL for social stimuli. Nature
- **Deep reinforcement learning can promote sustainable human cooperation in social dilemmas** (Koster et al., 2025): Recent RL social planner for trust games and sustainable contributions. Nature